

Analysis of Yelp Customer Reviews

Presented by Data Mavericks

Kashyap Mehta

Lin-Hsuan Chen

Pin-Lin Lai

Meshvee Kanani

Narayanan TK

Quick Walkthrough

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EDA

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Uncovering trends & patterns
Key findings and strategic recommendations

01

Problem Statement

What are we trying to solve?

Identifying Common Pain Points

Understanding critical factors leading to negative restaurant reviews on Yelp.

Customer Dislikes & Preferences

Investigating common themes in low-rated reviews to pinpoint what customers consistently dislike and prefer.

Impact on Business Success

Exploring how certain practices and offerings negatively affect restaurant ratings and customer satisfaction.

02

Project Overview

Let's move forward with California & Florida

We picked one state from the west and one from the east to conduct our analysis using the Yelp dataset

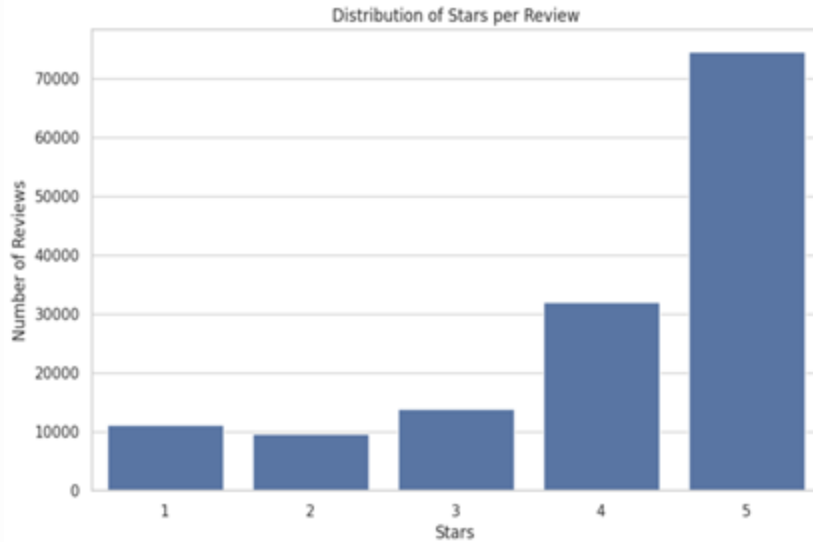


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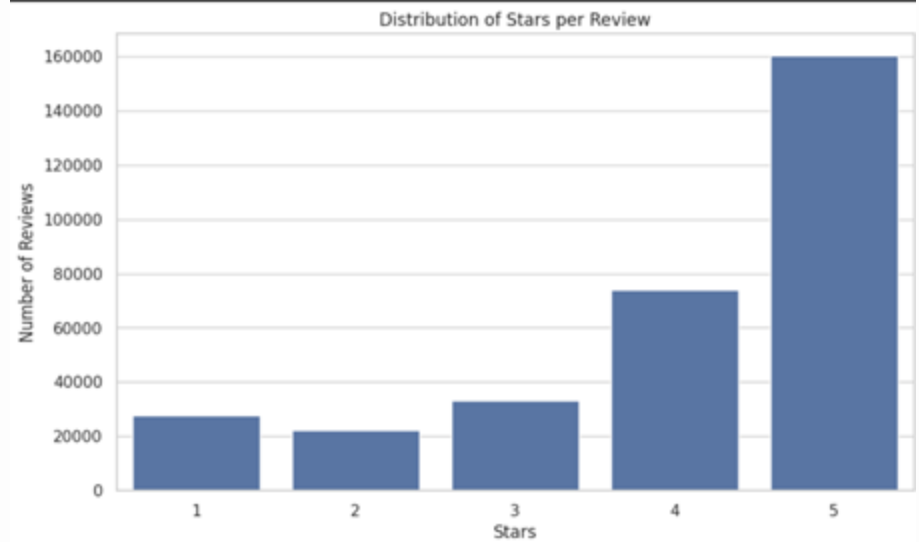
Exploratory Data Analysis

Review Distribution

Quick look at how many star reviews are there and what the distribution is like



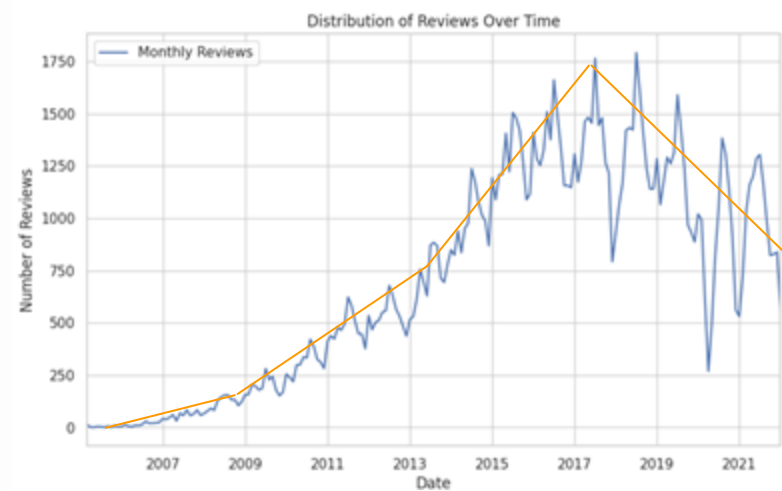
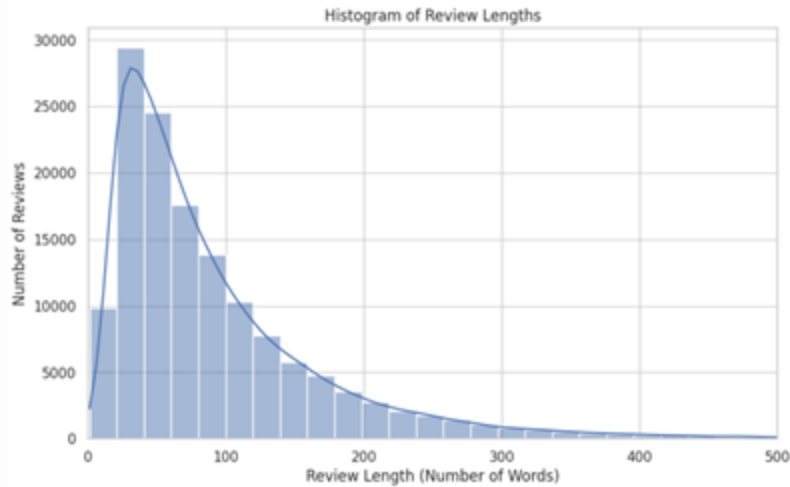
California



Florida

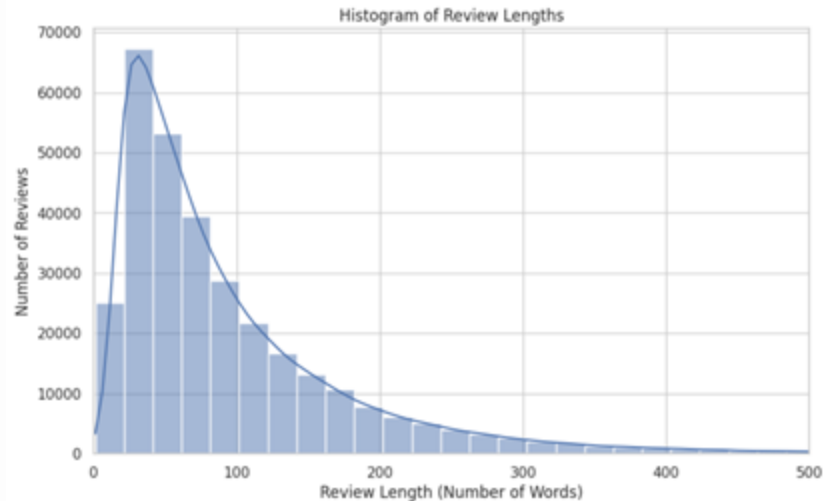
Some More Metrics - CA

Quick look at how many star reviews are there and what the distribution is like and how the number of reviews given by people has changed overtime in CA



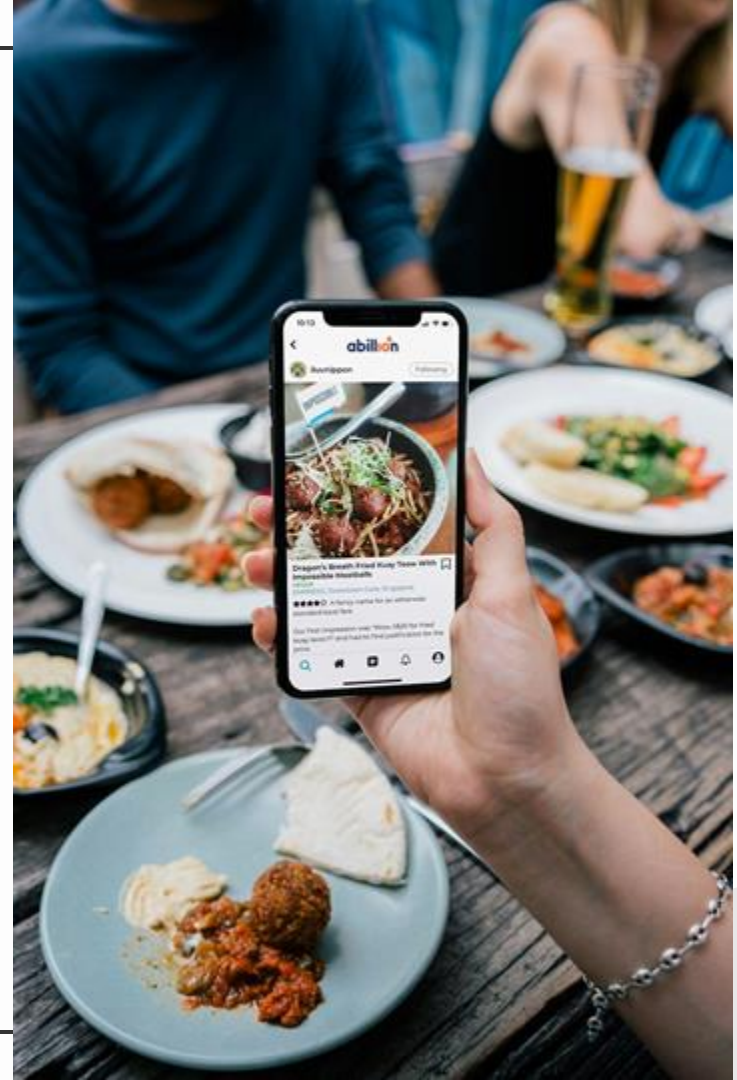
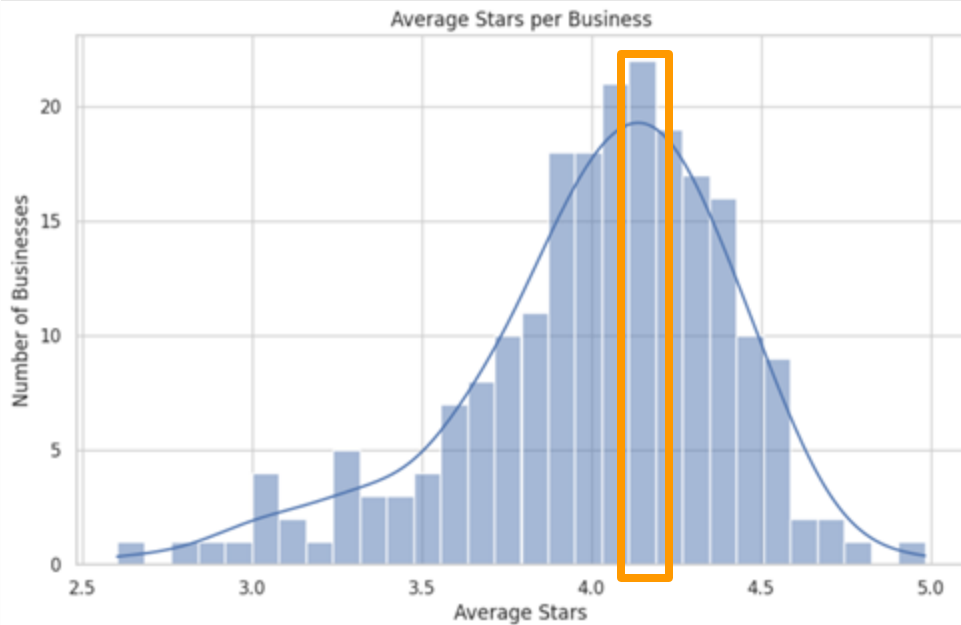
Some More Metrics - FL

Quick look at how many star reviews are there and what the distribution is like and how the number of reviews given by people has changed overtime in FL



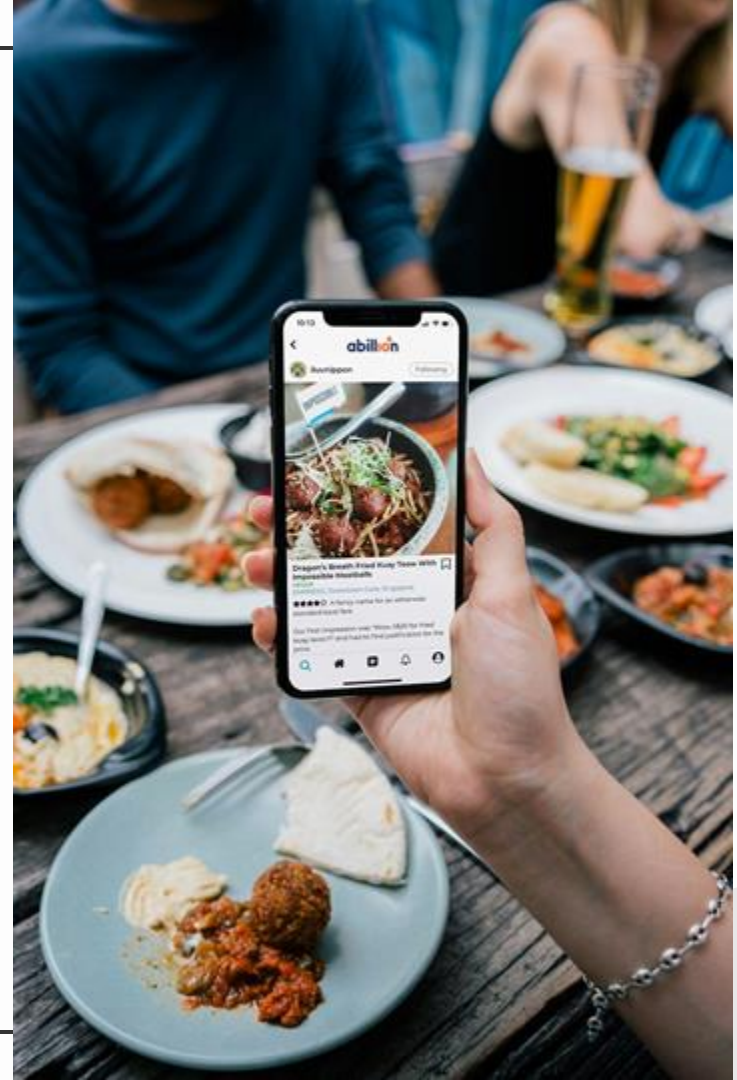
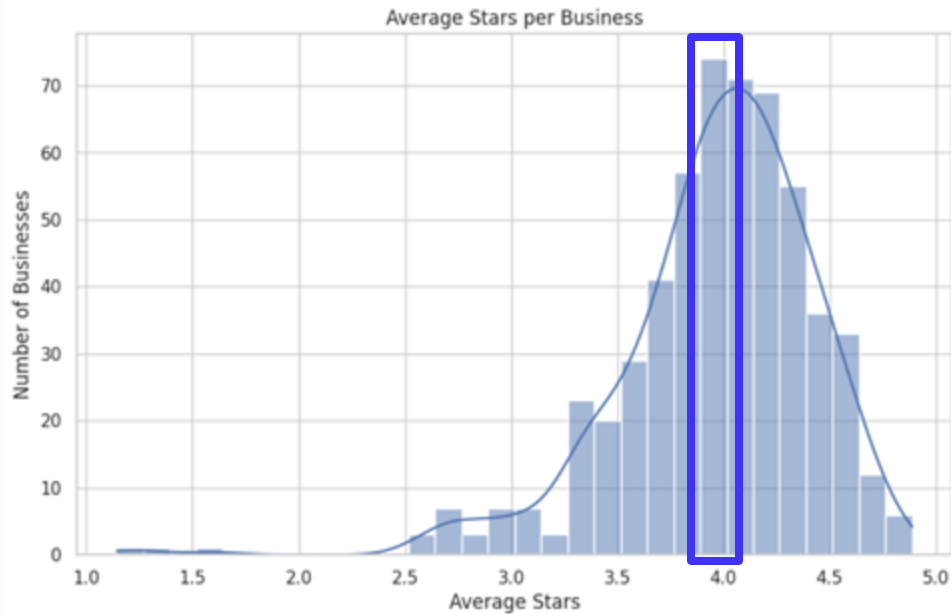
Business Ratings - CA

What ratings do people really give restaurants?



Business Ratings - FL

Is it the same trend in Florida as well?

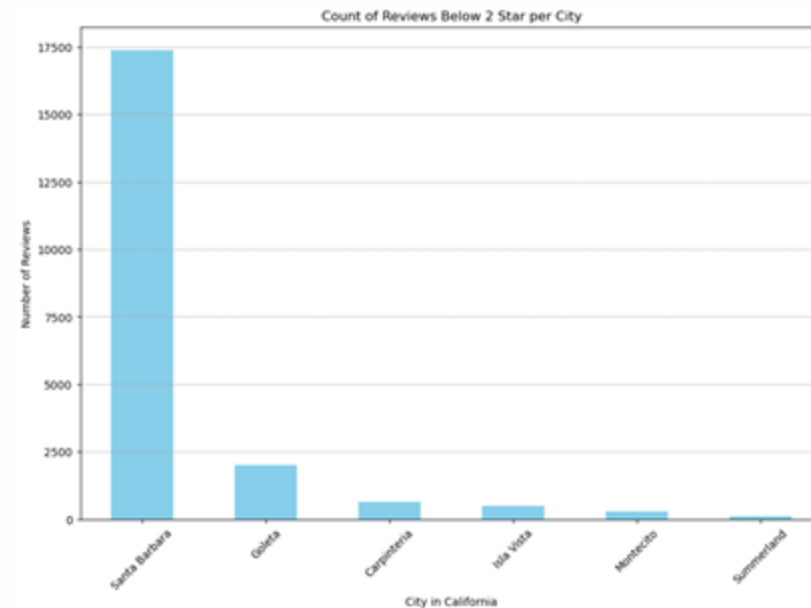
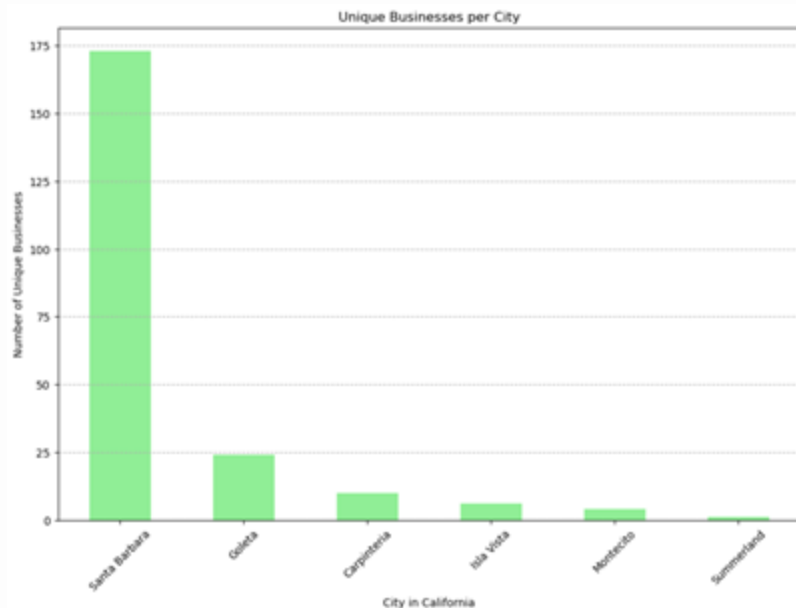


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Strategic Focus

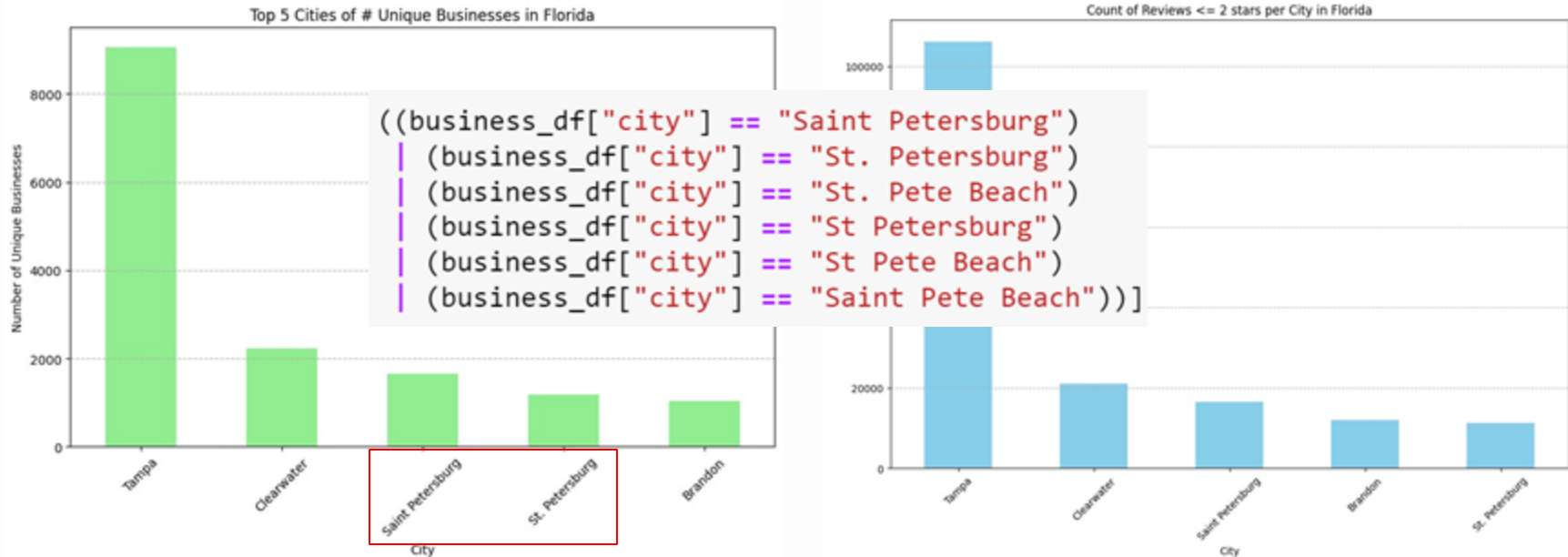
Narrowing Down Further in California

We wanted to check out the **top 3 cities** in each state in terms of number of businesses and number of reviews of restaurants that are **2 star rated and below** to conduct an unbiased analysis



Narrowing Down Further in Florida

We wanted to check out the **top 3 cities** in each state in terms of number of businesses and number of reviews of restaurants that are **2 star rated and below** to conduct an unbiased analysis



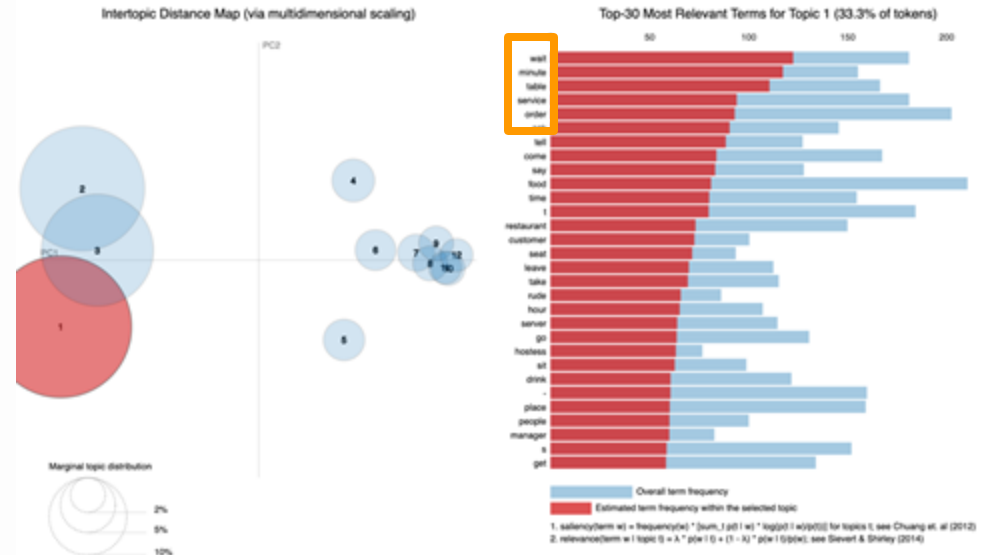
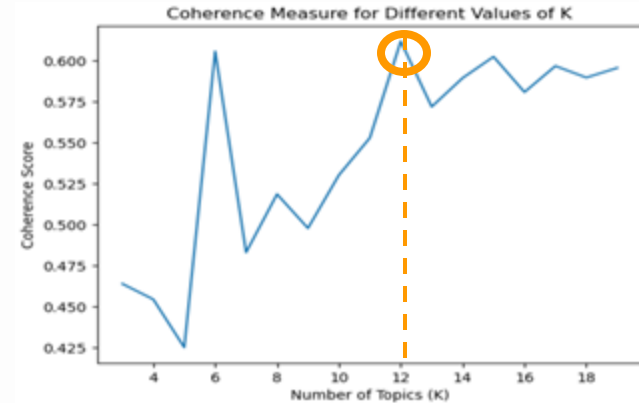
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Analysis & Conclusions

Santa Barbara - CA

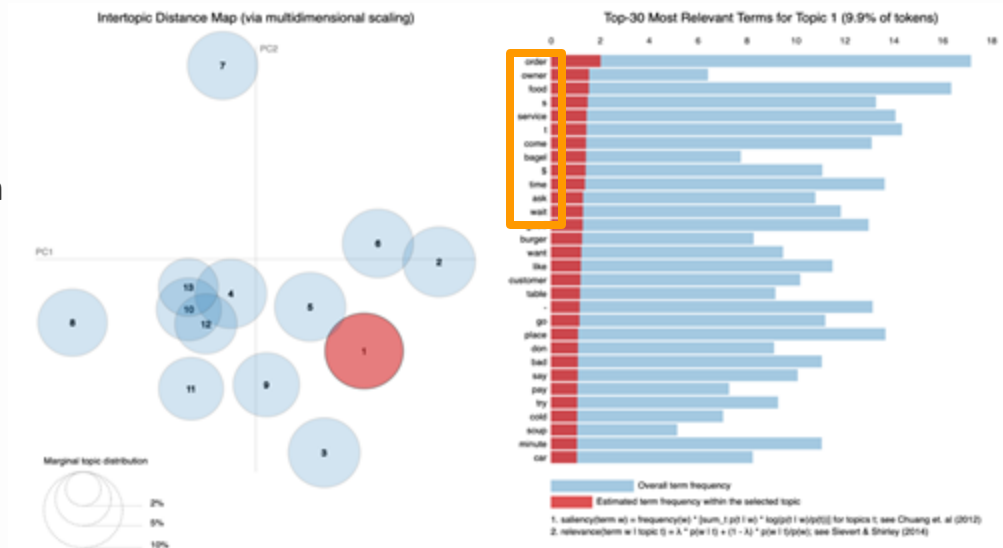
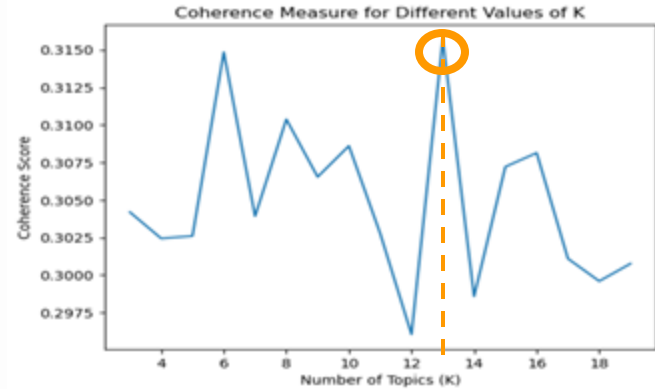
The analysis for the city of Santa Barbara showed these results

- The optimal value of K was 12 according to the coherence measure
- We observed that terms like “wait”, “minute”, “table”, “service” were one of the most frequently used terms



Goleta - CA

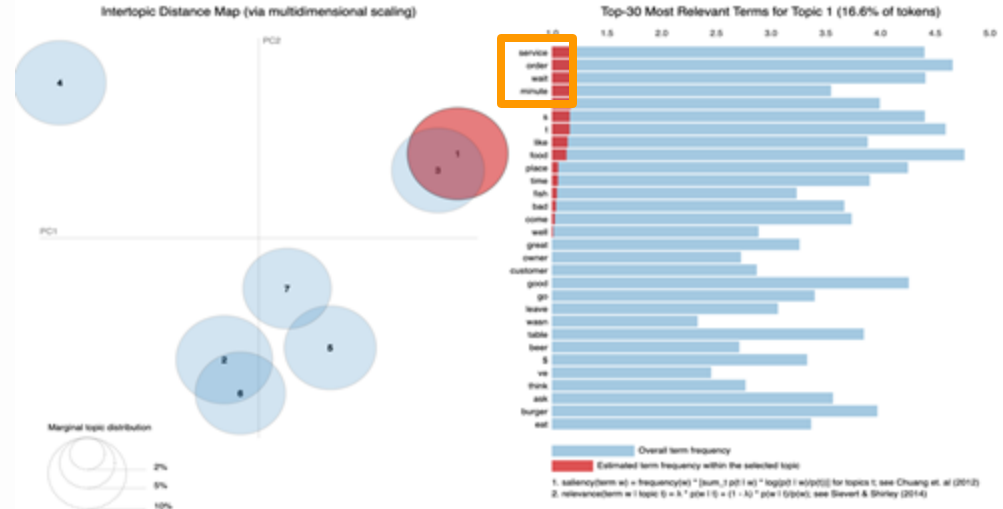
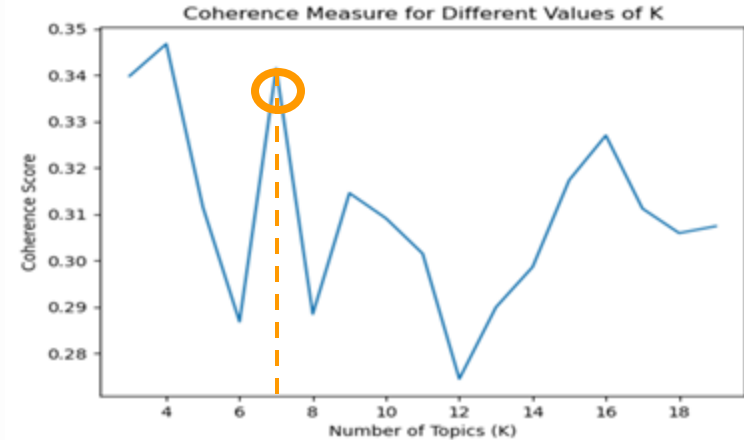
- students from nearby UC Santa Barbara, contributing to a demand for varied dining options, including Asian and specifically Japanese food
- student composition for UC Santa Barbara- White (42.1%) followed by Asian (32.2%) and Hispanic (18.3%).
- ASU- White 52%/ Black 4%/ Hispanic 19%/ Asian 7%



Carpinteria - CA

The analysis for the city of Carpinteria showed these results

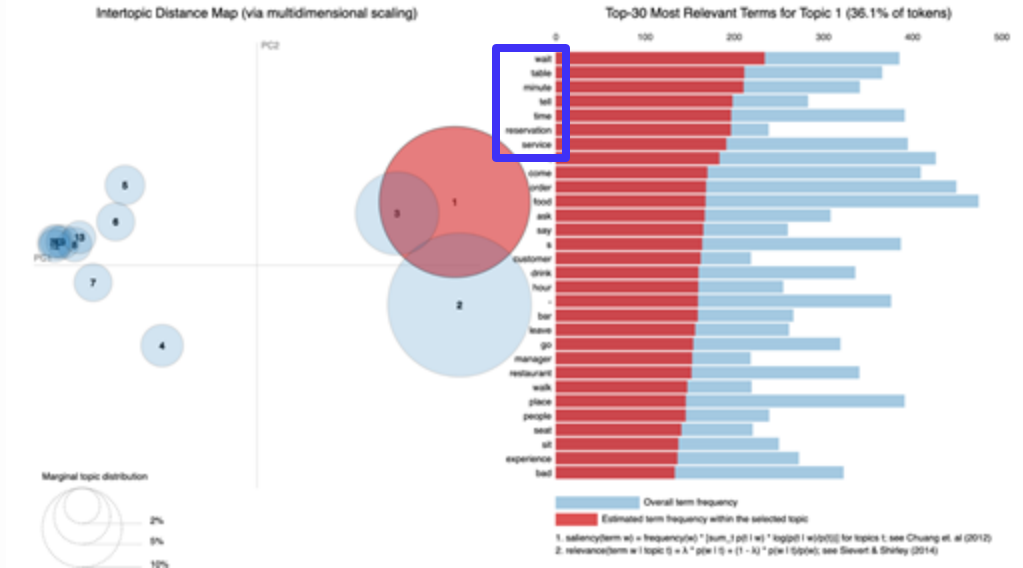
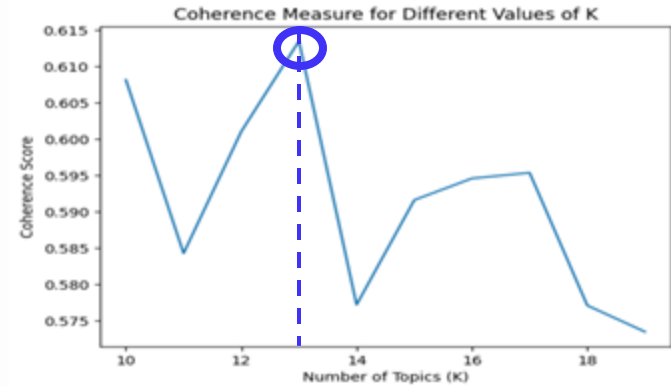
- The optimal value of K was 7 according to the coherence measure
- We observed that terms like “service”, “time”, “wait”, “minute” were one of the most frequently used terms
- We also observe a few terms like “food”, “good” used overall in a few topics



Tampa - FL

The analysis for the city of Tampa showed these results

- The optimal value of K was 13 according to the coherence measure
- We observed that terms like “reservation”, “time”, “wait”, “minute”, “table” were one of the most frequently used terms



St. Petersburg - FL

The analysis for the city of St. Petersburg showed these results

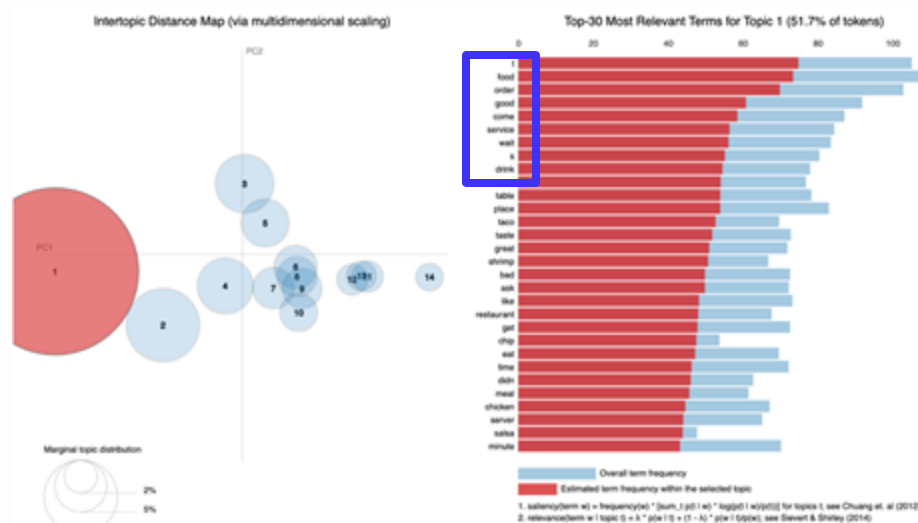
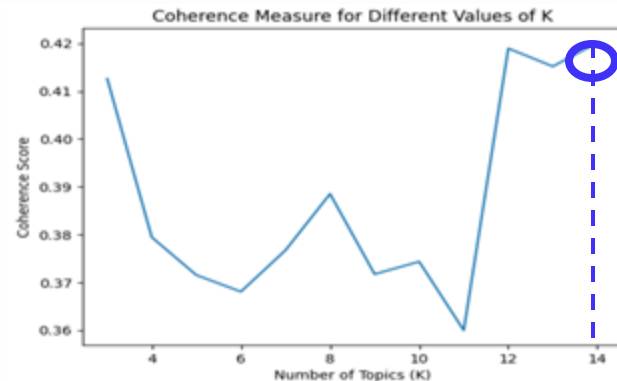
- The optimal value of K was 9 according to the coherence measure
- We observed that terms like “drink”, “come”, “wait”, “service”, “minute”, “time”, “order” were one of the most frequently used terms
- We also observe a few terms like “food”, “good”



Clearwater - FL

The analysis for the city of Clearwater showed these results

- The optimal value of K was 14 according to the coherence measure
- We observed that terms like “drink”, “come”, “wait”, “service”, “minute”, “time”, “order” were one of the most frequently used terms
- We also observe a few terms like “food”, “good”



One more interesting observation

- Though the overall star rating of restaurants were bad, their business reviews were good

business_id	review_id	user_id	stars_x	useful	funny	cool	text	date	name	city	stars_y	review_count
18eWJFJbXyR9j_5xfcRLYA	7ZF0NRbUgS2NnWitV1hoig	vmct6SMzG2vP-L15mmxG3A	1	0	0	0	Decent food but have some weird priorities and...	2017-05-13 04:08:42	Siam Elephant	Carpinteria	4.5	460
18eWJFJbXyR9j_5xfcRLYA	W914YNjoF9NEFyQcjup9bQ	cxmXMKlaDgbkpggh3ai-n0A	2	0	0	0	average Thai food. nothing spectacular peanut ...	2010-10-07 20:25:50	Siam Elephant	Carpinteria	4.5	460
18eWJFJbXyR9j_5xfcRLYA	F-yYtX93s9yQOdYVUbMNOw	Ue7CQEYntvEIORCNH5hijA	2	0	0	0	Service was great - really nice. Unfortunatel...	2012-06-27 21:15:58	Siam Elephant	Carpinteria	4.5	460
18eWJFJbXyR9j_5xfcRLYA	2lgzGI-d_Hz8aiSqKiEZtA	dpidfZbryC8D4hS0NYYOxQ	1	1	0	0	Walked into Siam Elephant with great expectati...	2016-02-26 05:00:02	Siam Elephant	Carpinteria	4.5	460
18eWJFJbXyR9j_5xfcRLYA	aPpfXHRHbvrsP96sSoHooQ	LZ9xjXeIM6tL4pCENoMmfg	1	0	0	0	This is the only Thai restaurant in Carpinteria...	2009-11-16 22:41:22	Siam Elephant	Carpinteria	4.5	460

CA Recommendations

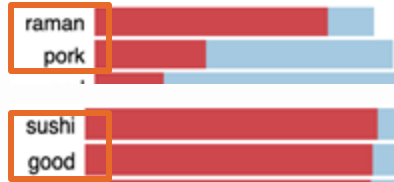
Santa Barbara

- History of burger chains doing well, like The Habit Burger Grill, showing there's a strong market for these food. which was founded in 1969 at Santa Barbara, has since expanded to over 300 locations.



Goleta

- Students from nearby UC Santa Barbara, contributing to a demand for varied dining options, including Asian and specifically Japanese food
- Student composition for UC Santa Barbara- Asian (32.2%)



Carpinteria

- Local wineries like Rincon Mountain Wine and Shokrian Vineyard are important; they make great wine and offer fun tasting experiences close to home. Carpinteria Wine Company also makes it easy and affordable for people to buy good wine.



FL Recommendations

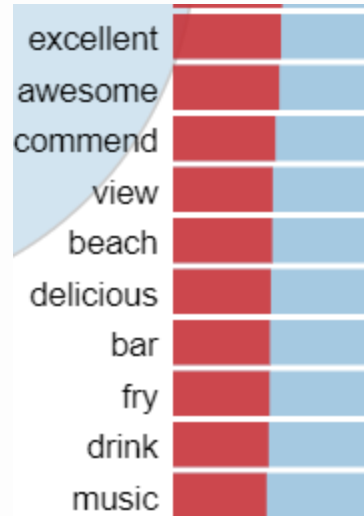
Tampa

- Looks like a lot of people like watching sport in sports bars according to our analysis.



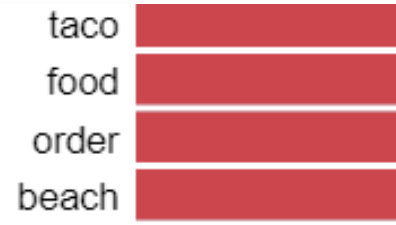
St. Petersburg

- Film festivals, live music, and vibrant arts scene, which likely includes a variety of entertainment options, contributing to a lively bar scene.



Clearwater

- The presence of many taco stores in Clearwater can indeed be influenced by the demographic composition of the area, including a significant Mexican or Latino population.





Thank You!

Questions?